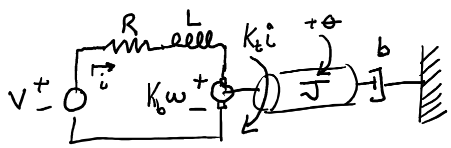
Lab 2 - Best to date F24 S5

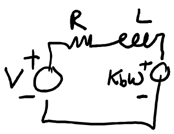


Electromechanical:

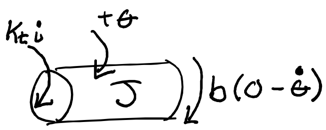
K_b = back EMF

K_t = torque constant

Electric Circuit



FBD



$$\left.\begin{aligned} V - Ri - L\frac{di}{dt} - K_b \omega &= 0 \quad \text{(KVL)} \\ K_t i - b\omega &= J\dot{\omega} \quad \text{(Newton)} \end{aligned}\right\} \text{ODEs}$$

eqns: 2

Knowns: Parameters: R, L, K_b, K_t, b, J

Input: V

2 unknowns: i, ω

Solve: numerically (LAB 1)

You! Laplace transform

$\mathcal{L}$

$$V - RI - LSI - K_b W = 0 \quad ①$$

$$K_t I - bW = JSW \quad ②$$

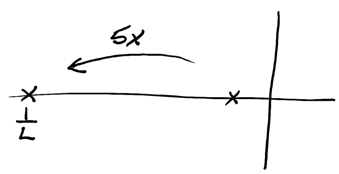
2 eqns.

2 unknowns: I, W

Find TF: $\frac{\omega}{V}$

$$\begin{array}{c} ① \\ ② \end{array} \begin{bmatrix} (R+LS) & K_b \\ K_t & -(JS+b) \end{bmatrix} \begin{bmatrix} I \\ W \end{bmatrix} = \begin{bmatrix} V \\ 0 \end{bmatrix} \rightarrow \text{Matlab}$$

$$\quad A \qquad\qquad x \qquad b$$

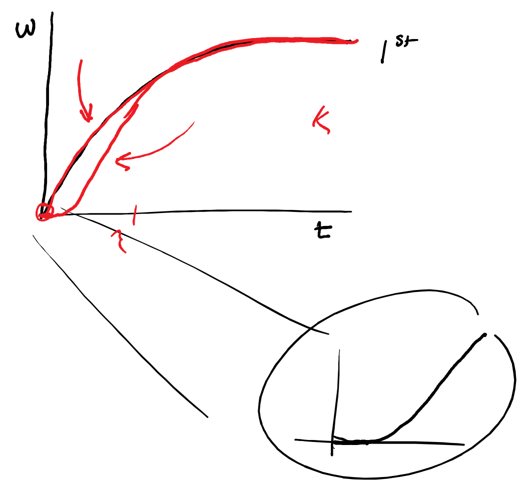


Assume inductance is Negligible $L=0$

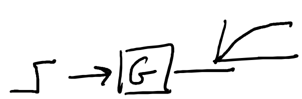
$$\frac{\omega}{V} = \frac{K_t}{J\cancel{L}s^2 + (JR + \cancel{bL})s + (Rb + K_t K_b)}$$

$$\boxed{\frac{\omega}{V} = \frac{K_t}{JRs + (Rb + K_t K_b)}} = \frac{1/(Rb+K_tK_b)}{1/(Rb+K_tK_b)} = \frac{K_t/(Rb+K_tK_b)}{JR/(Rb+K_tK_b)\, S + 1} = \boxed{\frac{K}{\tau s + 1}}$$

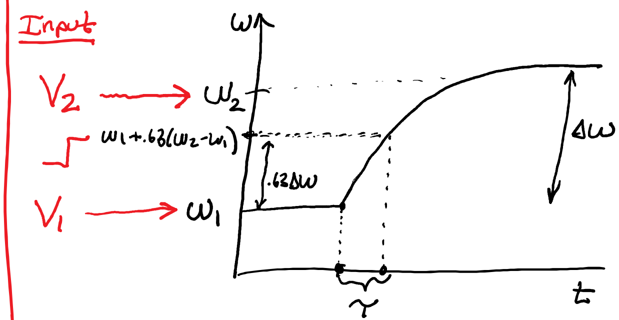
$$\boxed{\begin{aligned} K &= K_t/(Rb + K_t K_b) \\ \tau &= JR/(Rb + K_t K_b) \end{aligned}}$$



1^{st} order ODE/TF step Response



Non unit input, Non Zero output



$$K = \frac{\Delta \text{output}}{\Delta \text{input}} = \frac{\omega_2 - \omega_1}{V_2 - V_1}$$

Determining Motor Parameters: Steady-state Equations

$$V - Ri - \cancel{L\frac{di}{dt}}^{\,0} - K_b\omega = 0$$

$$V_{ss} - Ri_{ss} - K_b\omega_{ss} = 0$$

① • Apply a Voltage
• hold motor shaft $\omega_{ss} = 0$

$V_{ss} - Ri_{ss} = 0$ $\boxed{R = \frac{V_{ss}}{i_{ss}}}$ Equation for "stalled" motor

② • Apply Voltage
• Let motor spin freely

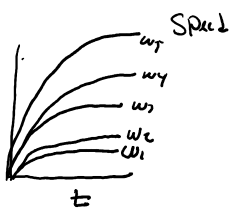
$\boxed{V_{ss} - Ri_{ss} - K_b\omega_{ss} = 0}$ equation for free-spinning motor.

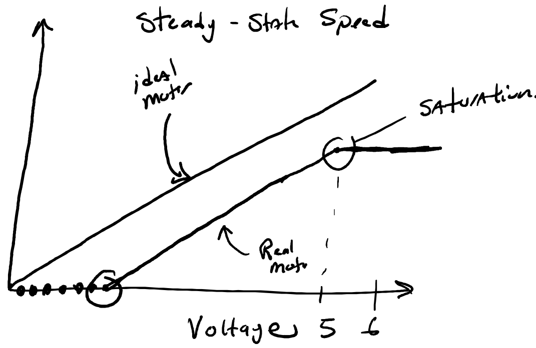
— Calculate K_b

ideal motor

Volts

• 1
• 2
• .5
1
3

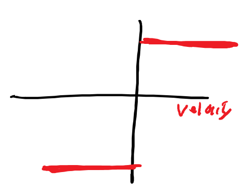




Mechanical Equation:

$K_t i - b\omega = J\dot{\omega}$ (ideal motor)

$K_t i - b\omega - \text{Coulomb} = J\dot{\omega}$
"Real" motor



$- \text{Sign}(v) \times \text{Coulomb}$